

Simple Line-Following Robot



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Generally, line-follower robots are microcontroller-based. Here we describe a line-follower robot without microcontroller

TCRT5000L reflective optical sensors (IC1 and IC2), inverter 74LS04 (IC3), motor driver L293D (IC4), regulator 7805 (IC5) and a few discrete components.

TCRT5000L optical sensors are

with sensors and controls the 12V geared DC motors M1 and M2. The L293D is a quadruple high-current, half-H driver. It is designed to provide bidirectional drive current of up to 600 mA at 4.5V to 36V. The device is designed to drive inductive loads such as motors as well as other high-current/high-voltage loads in positive-supply applications.

The motor drivers are enabled in pairs. When enable input pins 1 and 9 of IC4 are high, the associated drivers for M1 and M2 are enabled and their outputs pin 3 and 6, and pins 11 and 14, respectively, are active and in phase with the inputs.

With the proper data inputs, each pair of drivers forms an H-bridge reversible drive suitable for motor applications. Motors M1 and M2 rotate in forward direction (say clockwise) when both the sensor outputs are low.

If you want the robot to follow a black track (made using black-colour tape) on a white floor, you need to connect jumpers SJ1 and SJ2 to B terminals as shown in Fig. 1. The robot will move as follows: When the left sensor, say IC2, detects the black track, the robot

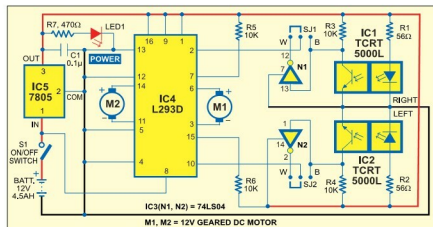


Fig. 1: Circuit of the line-following robot

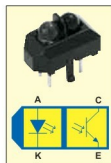


Fig. 2: Sensor and its internal details

for those who are not familiar with microcontrollers. It is a simple project, which can be taken up as a classroom assignment. It lays the foundation for building your own behaviour-based

(simulated) robot. The robot uses interfaces for the sensors to make the behaviour of the robot as versatile as possible. Two light detectors are mounted at the front of the robot. Also, a path is provided for the robot to follow – either a black track (using black colour tape) on a white floor or a white track on a black floor.

Circuit description

Fig. 1 shows the circuit of the line-following robot. It comprises two

used as line-detecting sensors. The TCRT5000L reflective sensor includes an infrared emitter and photo-transistor in a leaded package. The sensor and its internal details are shown in Fig. 2.

When an object comes in the sensing area, the emitted IR light reflects off the object back to the photo-transistor. So the amount of light energy reaching the detector increases. This change in light energy or photo-current is used as the input signal to activate the motors of the line-follower robot.

The collector current of the photo-transistor is also dependent on the distance of the object's reflector surface from the sensor. The intensity of the reflected signal returning to the detector depends on the surface over which the robot moves. White surface reflects maximum infrared signal, while black surface absorbs maximum infrared signal.

Motor driver L293D is interfaced

PARTS LIST

Semiconductors:

- IC1, IC2 - TCRT5000L reflector module
- IC3 - 74LS04 hex inverter
- IC4 - L293D motor driver
- IC5 - 7805, 5V regulator
- LED1 - 5mm LED
- DIS1-DIS5 - LT1543, CC7 segment display

Resistors (all ¼-watt, ±5% carbon):

- R1, R2 - 56-ohm
- R3-R6 - 10-kilo-ohm
- R7 - 470-ohm
- R8-R29 - 330-ohm

Capacitors:

- C1 - 0.1µF ceramic disk

Miscellaneous:

- M1, M2 - 25-rpm geared DC motor
- BATT. - 12V, 4.5Ah battery
- S1, S2 - On/off switch